



Patients self-mastery of wearable devices for seizure detection: A direct user-experience

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ABSTRACT

Purpose: wearable devices aimed at detecting seizures are rapidly emerging. Continuous collection and optimal data quality are paramount to guarantee the acquisition of clinically meaningful data. It is still unknown how successfully patients can self-manage new technologies and which factors have an impact on this. We assessed the performance of patients managing a wrist-worn device.

Method: patients wearing a wrist-worn device received a single training session to perform 5 tasks: (1) fitting the device correctly; (2) switching the device on and off; (3) charging the device on a daily basis; (4) pairing the device with a phone or tablet; (5) seeking assistance. Participants were then reviewed every 24 h and, at the end of the study, a Wearable technology Self-management Score (WSS) was calculated according to their performance in the different tasks (0–12). The association between WSS, seizure capture, demographics and clinical characteristics was analysed.

Results: Thirty patients were included. The mean WSS score was 9.4 ± 2.1 points. The task more often performed inaccurately was pairing the device with a phone or tablet, followed by performing charging procedures. A strong association was found between WSS and seizure capture ($p = 0.019$), with higher scores strongly associated with more seizures captured. A WSS of ≥ 9 was the minimum value of WSS that allowed the device to record all the seizures. Number of AEDs and illness-perception related factors (perceived disease timeline and personal control) were significantly associated with WSS.

Conclusions: Overall, participants demonstrated good performances in self-managing a wrist-worn device. Digital inequalities may extend to variations in how different individuals feel about their own disease and, consequently, manage the technology. These aspects should be addressed when technological solutions are delivered to users.

1. Introduction

Digital innovations applied to health monitoring are rapidly flourishing as the number of people owning wearable technology continues to rise [1]. A considerable body of literature has focused on wearable devices for seizure identification, supporting the development and marketing of a number of seizure detectors [2]. People with epilepsy (PWE) have expressed a strong interest in using new technologies in their daily life [3,4] and have identified a number of unmet needs that might be supplemented by adopting digital solutions into healthcare

services [5–8]. In addition to the key requirements of a reliable and accurate performance [2], a successful integration of digital solutions into a patient pathway has two major and interrelated requirements: independent use of the technology by PWE and long-term engagement. Health inequalities and exclusion may arise for specific groups of PWE according to the different degree of independence and confidence with the use of digital solutions. Prior experience with technology, patients' characteristics as well as disease-specific characteristics may represent a source of digital inequalities, acting as barriers to the use of technology [5,9–12]. However, it is still unknown whether such factors influence

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and prevent an optimal usage of the technology in real scenarios and, if so, to what extent. Moreover, despite the key role of technology self-mastery, data on how successfully PWE can self-manage new technologies have been collected only sporadically during direct user experiences [13–15]. The provision of technical support throughout patients' experience is an important aspect to increase device usability [3,5,12] with a few caveats: technical teams may not be easily identifiable and accessible and limited resources may limit addressing potential malfunctions in real time. We anticipate that an early identification of key aspects that could compromise technology usage would guide and inform a sensible allocation of resources in hospital and home settings, facilitate patient-device interaction and enable an optimal data acquisition.

In this study we assessed the self-mastery performance and associated factors in a group of PWE admitted to an epilepsy monitoring unit (EMU) who agreed to wear a wrist-worn device aimed at seizure detection.

2. Methods

2.1. Study participants

The current study has been developed in the context of RADAR-epilepsy, a multicenter study designed to assess the clinical utility of multiparametric remote measurement technologies (RMT) in a clinical population with epilepsy, in the hospital and real-world environment [16]. The study population consisted of consecutive PWE who were admitted, for diagnostic reasons or presurgical evaluation, to the EMU at King's College Hospital, London. For the scope of this study participants admitted to the EMU for at least 3 consecutive days were considered and included.

2.2. Ethics approval

The trial and study procedures were approved by the London Fulham Research Ethics Committee (16/LO/2209; IRAS project ID216316). All participants provided written informed consent.

2.3. Data collection

Participants were asked to wear a wrist-worn multimodal device (Empatica E4, Fig. 1) for the entire duration of their stay in the EMU.

The sensors included in this wearable and the parameters measured are photoplethysmography (PPG) and series of interbeat intervals derived from it, 3D acceleration, body temperature and electrodermal activity (EDA). The device was worn on the non-dominant hand. For each patient, three categories of variables were collected: (1) patient-

related: demographic data, including age and gender, ethnicity, marital status, level of education, employment, past experience with wearable technologies; (2) epilepsy-related: age at onset of epilepsy, disease duration, type of epilepsy, seizure frequency, medical comorbidities, mental health comorbidities (depression and anxiety), number of anti-epileptic drugs (AEDs), number of other medications, total number of medications; (3) illness perception-related, using the Brief Illness Perceptions Questionnaire (BIPQ) [17]. The questionnaire includes 8 dimensions with a possible score from 0 to 10 points (0 not at all, 10 extremely): consequences and overall burden (How much does your epilepsy affect your life?), perceived disease timeline (How long do you think your epilepsy will continue?), personal control (How much control do you feel you have over your epilepsy?), treatment control (How much do you think your treatment can help your epilepsy?), perceived severity (How much do you experience symptoms from your epilepsy?), concerns (How concerned are you about your epilepsy?), understanding (How well do you feel you understand your epilepsy?) and emotional response (How much does your epilepsy affect you emotionally?). The BIPQ total score was calculated as the sum of the reversed scores of items 3, 4, and 7 and the scores of items 1, 2, 5, 6, and 8 (maximum score 80 points). A higher BIPQ total score indicates that the patient views the condition as more burdensome.

At the end of the study period, the participants' experience and the perceived ease of use of the technology was assessed using a self-administered questionnaire: the Technology Acceptance Model Fast Form (TAM-FF) [18]. Participants scored the following subjects on a 7-point Likert-scale (1, strongly agree to 7, strongly disagree): Overall, I feel the wearable device is (1) "easy to manipulate (wear/fit)"; (2) "easy to interact with (switch on/off)"; (3) "easy to master (e.g. charging)"; (4) "flexible to interact with (easy to pair)"; (5) "easy to learn (no assistance needed)"; (6) "usable (independent use)".

2.4. Training, self-management tasks and assessment

On day 1 (study enrollment), participants received a 15-minute training session on the use of the wearable device. In particular, the training was focused on how to perform the following key tasks: (1) fitting the device correctly; (2) switching the device on and off; (3) charging the device on a daily basis (and swap with a spare device every 12 h during charging procedures); (4) pairing/re-pairing the device with a phone or tablet (the device was paired via an ad-hoc created app running on a phone or tablet. The app indicated when the device got unpaired due to loss of Bluetooth connection. The task consisted in repair the device each time it accidentally got unpaired due to loss of connection as indicated by the app on the phone or tablet); (5) seeking assistance should any problem arise (for any of the above-mentioned tasks or for additional technical problems). At the end of the training session, study participants were asked to start wearing the device and the recording started. Participants were then reviewed every 24 h from day 2 onwards, until the day of discharge from EMU (study endpoint). During each daily assessment, the research team annotated whether each task was performed correctly or not (yes or no), using an electronic form. Additional training and help were offered when a task was performed incorrectly at any point during the study period and the need for assistance was recorded on the electronic form on a daily basis. At the end of the study, a score of 2 was assigned to each task when the task was always performed correctly from day 2 onwards; a score of 1 when the task was performed correctly for $\geq 50\%$ of the days and of 0 when performed correctly for less than 50% of the time. An additional score of 2 was assigned when participants were "independent" and did not require any additional training; a score of 1 when "aided" (re-training required on one or more tasks) and a score of 0 when participants required "constant support" in performing some or all the tasks. A Wearable-technology Self-management Score (WSS) was then calculated as the sum of these 6 scores (0 lowest performance to 12, highest performance).



Fig. 1. Empatica E4 wristband.

A seizure was defined as “captured” when the time window during which the seizure occurred, established on the basis of the video-EEG, was recorded by the device, independently from the capacity of the device to automatically detect the seizure, which is not part of the scope of this research. The total number of seizures presented by each participant during the study period and the number of seizures not recorded by the device due to self-management tasks performed incorrectly was documented. The binary variable “seizure capture” (at least one seizure lost *versus* all seizures recorded) was created for each participant.

2.5. Statistical analysis

As seizure capture is paramount to allow an effective seizure detection (which is the main goal of seizure detection devices), the association between the WSS and the binary variable “seizure capture” during the study period, was analysed using logistic regression. The minimum value of WSS that allowed the device to record all the seizures presented by each participant was chosen as the WSS cut-off indicating an optimal self-management. Conversely, values below the cut-off were considered as indicating poor self-management (at least one seizure lost). To identify the WSS cut-off value, the sensitivity and specificity of each WSS score was calculated using a ROC analysis. The area under the curve (AUC) was also obtained. Cronbach’s alpha was used to assess the WSS internal consistency. The association between patient-related, epilepsy-related, illness perception-related factors, days spent in the study (study duration) and WSS was analysed as follows: Pearson correlation and correlation coefficient were used to analyse the relationship between the WSS score and other continuous variables and to quantify the strength and direction of the relationship between the different pairs of variables; a two-sided Mann-Whitney-Wilcoxon (MWW) rank-sum test was used for binary variables whilst the Kruskal-Wallis (KW) test was used for categorical variables. Each test was performed at a significance level of 0.05. Data were processed using STATA version 14.0 (StataCorp, College Station, TX, U.S.A.).

3. Results

3.1. Study participants

Thirty consecutive participants admitted to the EMU between June 2017 and March 2020 were asked to wear and self-manage the wrist-worn device. Demographic and clinical characteristics are reported in Table 1. The cohort mainly comprised patients admitted for pre-surgical evaluation of their pharmaco-resistant epilepsy. The majority had a diagnosis of temporal lobe epilepsy (53.4 %), were on polytherapy (median number of AEDs = 3) with a long epilepsy duration (17.9 ± 12.9 years) and frequent seizures (11.6 ± 11.8 seizures/month). Only 30 % had previous experience with wearable technology. The mean duration of participation in the study was 6.0 ± 2.3 days during which 96 seizures occurred. The device recorded a total of 4,128 h across all participants.

3.2. WSS score

The mean WSS score was 9.4 ± 2.1 points. How participants scored on each task is summarized in Fig. 2.

Overall, the vast majority of participants managed to wear and fit the device correctly (90.0 %), to switch it on and off (70.0 %) and to charge it (63.3 %). The task more often performed inaccurately was pairing the device with a phone or tablet, followed by performing charging procedures and swapping with a spare device. The latter was more often due to participants forgetting to charge and swap the device, rather than to an incorrect procedure. Up to 50.0 % of our cohort was entirely independent in performing all the tasks, 36.7 % required additional support/training and 13.3 % needed a constant supervision in some or all the

Table 1

Participants characteristic and association between WSS score and patient-related, epilepsy-related and illness-perception-related variables (N = 30).

Variables	N (%)	Analysis	
		r	p-value
Patient-related variables			
Gender			0.06 (MWW)
Female	16 (53.3)		
Male	14 (46.7)		
Age (years) mean ± SD	40.9 ± 14.1	−0.01	0.9
Ethnicity			0.3 (KW)
White	22 (73.3)		
Black/African/Caribbean/Black British	5 (16.7)		
Asian/Other	3 (10.0)		
Marital status			0.2 (KW)
single	10 (33.3)		
married	13 (43.4)		
cohabiting with partner	3 (10.0)		
divorced	3 (10.0)		
widowed	1 (3.3)		
Level of education			0.2 (KW)
Secondary school	10 (33.3)		
Undergraduate degree	15 (50.0)		
Postgraduate degree (University)	5 (16.7)		
Employment			0.4 (MWW)
employed	16 (53.3)		
unemployed	14 (46.7)		
Experience with wearables			0.7 (MWW)
yes	9 (30.0)		
no	21 (70.0)		
Study duration (days) mean ± SD	6.0 ± 2.3	−0.3	0.3
Epilepsy-related variables			
Age at epilepsy onset (years) mean ± SD	23.0 ± 16.3	0.00	0.9
Epilepsy duration (years) mean ± SD	17.9 ± 12.9	−0.03	0.9
Epilepsy type			0.4
Temporal	16 (53.4)		
Extratemporal	6 (20.0)		
IGE	2 (6.6)		
Unknown	6 (20.0)		
Seizure frequency per month mean ± SD	11.6 ± 11.8	−0.06	0.7
Medical comorbidities			0.6 (MWW)
present	19 (63.3)		
absent	11 (36.7)		
Mental health comorbidities			0.2 (MWW)
present	8 (26.7)		
absent	22 (73.3)		
Number of AEDs median (range)	3 (1–5)	0.3	0.05
Number of other medications median (range)	1 (0–7)	−0.08	0.6
Total number of medications median (range)	3 (1–10)	0.1	0.6
Illness perception (BIPQ scores) mean ± SD			
Overall burden	7.5 ± 1.8	−0.06	0.8
Perceived disease timeline	8.5 ± 2.5	−0.4	0.05
Personal control	3.9 ± 3.5	0.4	0.05
Treatment control	6.9 ± 2.6	0.1	0.6
Perceived severity	6.2 ± 2.2	0.04	0.5
Concerns	7.7 ± 2.0	−0.1	0.5
Understanding	6.9 ± 2.6	0.2	0.2
Emotional response	7.9 ± 2.1	−0.3	0.1
BIPQ total score	49.6 ± 9.6	−0.4	0.03

AEDs: anti-epileptic drugs; BIPQ: Brief Illness Perceptions Questionnaire; IGE: idiopathic generalized epilepsy; KW: Kruskal-Wallis; MWW: Mann-Whitney-Wilcoxon; r: Pearson coefficient; SD: standard deviation.

tasks. The TAM-FF mean scores for each item are also reported in Fig. 2. TAM-FF mean scores for each subject ranged between 1.5 ± 0.8 and 2.2 ± 1.7 points (Fig. 2), indicating that overall the use of the technology was considered effortless. The highest mean score at the TAM-FF (2.2 ± 1.7 points), corresponding to a less positive judgment, was given to the

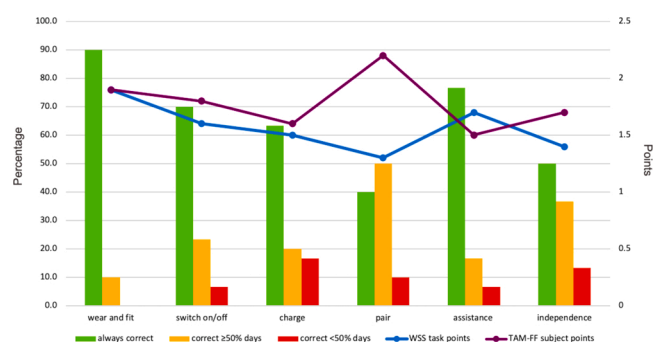


Fig. 2. Percentage of participants performing always correctly (green), correctly in $\geq 50\%$ of days (yellow) or correctly in $< 50\%$ of days (red) at each of the six WSS (Wearable technology Self-management Score) tasks assigned. Mean points obtained at each WSS task (0 to 2, blue line) and mean points at each TAM-FF (Technology Acceptance Model Fast Form) subject (1 to 7, purple line) (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

subject “flexibility to interact with the device”, corresponding to the WSS task “pairing the device with a phone or tablet” (Fig. 2). Due to sub-optimal performance, 19/96 seizures (19.8 %) were not captured by the device in eight participants, and 100 % of the seizure presented were lost in 5/8 participants. The logistic regression demonstrated a strong association between WSS and seizure capture ($p = 0.019$), with higher scores strongly associated with more seizures captured. The ROC analysis demonstrated that a WSS of ≥ 9 was the minimum value of WSS that allowed the device to record all the seizures, with a sensitivity of 100 % and a specificity of 87.5 %. The AUC for the WSS was 0.95, indicating that the score was excellent at discriminating optimal self-management (all seizures recorded) from poor self-management (at least one seizure lost). The Cronbach’s alpha for the WSS was 0.7, indicating an acceptable internal consistency.

3.3. Associated factors

Results of the analysis of the variables associated with the WSS are reported in Table 1. No statistically significant relationships were found for patient-related factors and number of days in the study. Among epilepsy-related factors, participants taking 3 ($N = 9$), 4 ($N = 4$) or 5 ($N = 2$) AEDs were found to have higher (> 9 points) WSS (better performance) (Pearson coefficient = 0.3; $p = 0.05$) (Fig. 3).

Illness-perception related factors were associated with WSS. In particular, participants with a higher BIPQ total score (higher disease burden) had lower WSS (worse performance) (Pearson coefficient = -0.4;

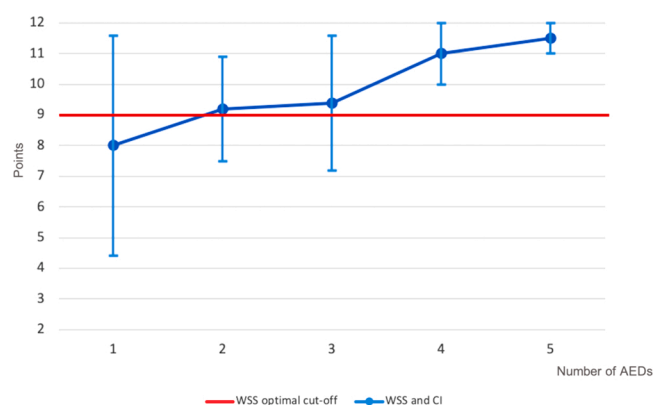


Fig. 3. Number of AEDs (antilepileptic drugs) and mean WSS (Wearable technology Self-management Score) with confidence intervals (CIs). Pearson coefficient = 0.3; $p = 0.05$.

$p = 0.03$) as well as participants with higher score at the dimension “perceived disease timeline” (higher burden) (Pearson coefficient = -0.4; $p = 0.05$) which had lower WSS (worse performance). Participants with lower scores at the dimension “personal control” (higher burden) had lower WSS (worse performance) (Pearson coefficient = 0.4; $p = 0.05$).

4. Discussion

In a cohort of mainly “device naïve” patients with uncontrolled severe epilepsy, our study demonstrated that many, but not all, participants had overall good device self-management skills and were able to set-up and operate our system effectively. Our study identified a number of factors that were associated with sub-optimal device management and incomplete seizure capture. At a time when wearable devices and mobile health apps are rapidly developing, patients are going to be entrusted with the responsibility to facilitate optimal data collection, and the evaluation of their response to this task is of primary importance. Only a few studies assessing wearables for epilepsy has explored patients’ experience, usually by collecting subjective feedback [4,6,13–15]. To the best of our knowledge, none of these studies has focused on a direct assessment of patients’ independence and there is very limited knowledge about specific factors potentially affecting confidence in using technologies. Previous investigations have demonstrated that PWE value the possibility to actively participate in their own health monitoring and consider such opportunity as empowering [3,5]. However, the same studies highlighted that technical support is required to guarantee and perpetuate autonomy and to avoid creating additional burden in a chronic condition.

In our study, a-priori identified key technical tasks were used to guide participants’ training and to objectively reassess and quantify their performance and autonomy. This approach allowed the development of a score system (WSS), which was strongly associated with seizure capture rate, and the identification of a cut-off WSS value to differentiate optimal self-management from poor self-management. At the same time, tasks at higher risk of error were pinpointed. In particular, pairing the device with a phone or tablet, charging and swapping the device were identified as critical steps in our setting. The same results were also reflected in the TAM-FF self-administered questionnaire, highlighting that participants were aware of their performance difficulties. Future studies using similarly operating devices should implement solutions to mitigate the risk of similar errors. Alarms and notifications alerting users to reconnect a device when unpaired or reminding them when it is time to charge may be of great assistance, as well as technical improvements of the technology (automatic pairing, wireless battery charging).

Our field study has also demonstrated that only half of the cohort of patients included was entirely independent in the first week of a new device use, whilst the other half needed assistance and technical support to different extents. This finding confirms the importance for many users of an easily identifiable and accessible technical support to improve device usability and to address potential malfunctions. However, despite the highly supervised setting, seizures were lost in some participants and were entirely lost in 5 of them. Engaging with new technologies could potentially add to inequalities and health disparities and this should be avoided by recognizing specific populations and groups who might be at higher risk of digital exclusion [5,9,12] or at risk of providing sub-optimal data. Evidence has indicated that patient-related factors, such as age [1,9,12] can influence interaction with new technologies (also in terms of technical support needed [12]), while other studies suggested that individuals with less education [19] and people with psychiatric comorbidities may be generally subject to poor outcomes [19–22]. In our population, technology self-management did not differ between younger and older participants, and no differences were found according to the level of education or to the presence of psychiatric comorbidities, suggesting that a tailored training could, at least in part, level the disparities that might otherwise occur. Similarly,

epilepsy-related factors, including disease duration and seizure frequency, did not affect participants performance, with the exception of number of AEDs taken. Participants taking 3 or more AEDs were found to have higher WSS, indicating better performance. The small numbers prevent us from a clear explanation of this result. Nevertheless, we may speculate that patients accustomed to follow a strict and complex drug regime may respond more appropriately to self-management tasks.

Unexpectedly, we found that the most important factors influencing self-mastery were related to the individual perception of the disease. The BIPQ indicated that participants considering their epilepsy as more burdensome had lower WSS, indicating worse performances. Specifically, the dimension “perceived disease timeline” and “personal control” significantly interfered with the tasks given, suggesting that a more negative view of the duration of the disease and of the possibility to control symptoms may be detrimental to optimal data acquisition. These findings may be related to the influence of the above mentioned factors on self-efficacy, a primary determinant of self-management of epilepsy and therapies [22–24]. The continued failure to control seizures and the long disease duration may lead to a form of perceived “defeat” with consequent low level of confidence and weak sense of efficacy, all influencing self-management performance [22]. As one of the potential target population who might benefit from seizure detection devices is represented by patients with uncontrolled seizures and many years of disease behind and ahead, these findings acquire a strong practical importance. In parallel with a more intensive form of training, additional approaches, including psychological and cognitive therapies aimed at reducing the perceived impact of epilepsy on quality of life and future perspective, might be of assistance in this group.

As an alternative explanation, although epilepsy severity-related factors, including disease duration and seizure frequency, did not affect participants performance, we cannot exclude that compromised executive function may have played a role in this finding. However, this was not specifically investigated due to the lack of neuropsychological data.

Finally, we would like to point-out some limitations of the current study. Firstly, due to the short follow-up duration and to the setting, questions related to the self-management performance in the long-term and in unsupervised environments remain unanswered. Secondly, the tasks included for the calculation of the WSS were device-specific and thus not immediately applicable to studies including a different device. However, the WSS provided a solid framework, strongly associated with seizure capture, that could be easily adapted for the majority of the devices currently on the market. Finally, as it is not just the individual users who are going to deal with the technology, the performance of those who are more likely to support and care for PWE should also be assessed in future studies. We should underline that the low correlation coefficients found in our analysis suggest that a non-linear relation may exist between some of our variables, and different models should be investigated in larger datasets.

Concluding, PWE appear to be ready to add an additional layer of self-management to their routine care. Variables such as number of AEDs and scores to specific sections of the BIPQ were found to influence the WSS score and may be easily obtained prior to recommending a device to predict performances. In fact, our data suggested that digital inequalities also extend to variations in how different individuals feel about their own disease and, consequently, engage, manage and cope with technology, and this aspect needs to be considered when technological solutions are delivered to users.

Declaration of Competing Interest

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